

## PROJECT DESIGN PHASE-II

### Technology Stack (Architecture & Stack)

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 01 October 2026                                |
| Team ID       | SWTID-2026-9397                                |
| Project Name  | Auto-Ticket Classification using Flow Designer |
| Maximum Marks | 4 Marks                                        |

#### Technical Architecture

The solution is implemented within the ServiceNow platform. The student uses the School IT Tickets interface, ticket records are stored in the custom School IT Ticket table, Flow Designer applies the classification rules, and ServiceNow email, ACLs, roles and Update Sets support notification, security and controlled change tracking.

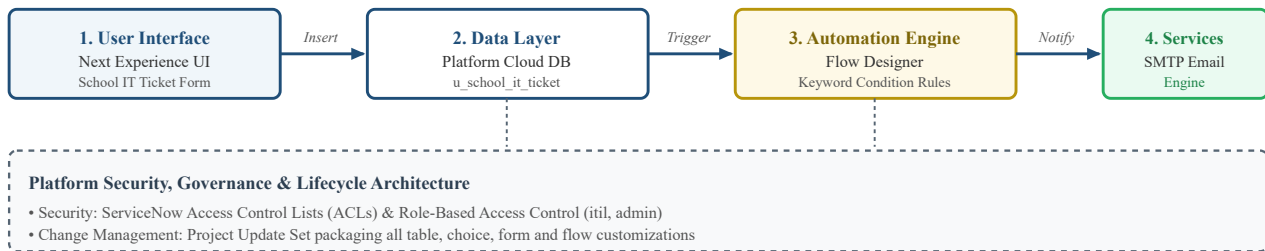


Figure 1: Technical architecture of the ServiceNow-based ticket automation solution.

#### Technical Architecture Diagram Explanation

##### System Architecture Breakdown:

- **Presentation Layer:** Students submit requests via the ServiceNow Next Experience School IT Tickets form. Technicians view records through standard list and form views.
- **Database Layer:** Ticket records are stored in the custom table `u_school_it_ticket`, with custom fields for short description, category, subcategory, state, and caller reference.
- **Automation Engine:** Flow Designer executes natively in the platform cloud without external middleware, evaluating condition branches on record insert.
- **Platform Services:** ServiceNow SMTP Email service sends automated notifications, ACLs govern record permissions, and Update Sets handle change management.

**Table-1: Components & Technologies**

| S.No | Component              | Description                                                          | Technology                   |
|------|------------------------|----------------------------------------------------------------------|------------------------------|
| 1    | User Interface         | Interface used to create and view School IT Tickets.                 | ServiceNow Next Experience   |
| 2    | Application Logic      | Automatic keyword-based classification and notification workflow.    | ServiceNow Flow Designer     |
| 3    | Database               | Stores School IT Ticket records and fields.                          | ServiceNow platform database |
| 4    | Cloud Database         | No separate cloud database is used by this solution.                 | Not used                     |
| 5    | File Storage           | No application-specific file storage is required.                    | Not used                     |
| 6    | External API           | No third-party API is required for the implemented workflow.         | Not used                     |
| 7    | Machine Learning Model | No machine-learning model is used; classifier is rule/keyword based. | Not used                     |
| 8    | Security               | Controls access to the custom table and records.                     | ServiceNow ACLs and Roles    |
| 9    | Change Tracking        | Tracks table, dictionary, choice, flow and related customizations.   | ServiceNow Update Sets       |
| 10   | Infrastructure         | Application execution and platform services.                         | ServiceNow Cloud Platform    |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                 | Technology                       |
|------|--------------------------|---------------------------------------------------------------------------------------------|----------------------------------|
| 1    | Open-Source Frameworks   | No separate open-source application framework is used in the implemented solution.          | ServiceNow platform              |
| 2    | Security Implementations | The custom ticket table uses access control and role configuration.                         | ServiceNow ACLs / Roles          |
| 3    | Scalable Architecture    | The event-driven flow can apply the same classification process to multiple ticket records. | Flow Designer                    |
| 4    | Availability             | The application is accessed through the ServiceNow platform and School IT Tickets module.   | ServiceNow platform              |
| 5    | Performance              | Classification occurs automatically after record creation for supported issue keywords.     | Flow Designer conditions/actions |

## Implementation Evidence

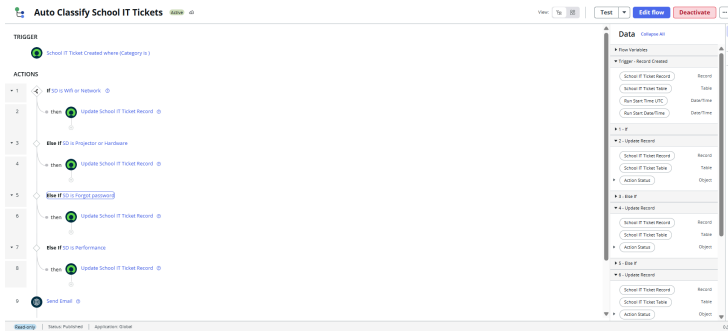


Figure 2: Published Flow Designer workflow.

The screenshot shows the ServiceNow "Update Set - Project Update Set" interface. The "Name" field is "Project Update Set", and the "State" is "Complete". The "Created" date is "2024-09-20 10:17:17". The "Created by" is "admin". The "Merged to" field is empty. The "Update Set" section shows a list of updates with columns: "Created", "Type", "View", "Target name", "Updated by", "Remote update set", and "Action".

| Created             | Type             | View | Target name                     | Updated by | Remote update set | Action           |
|---------------------|------------------|------|---------------------------------|------------|-------------------|------------------|
| 2024-09-20 10:17:17 | Application Menu |      | School IT Support               | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Module           |      | School IT Tickets               | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Choice list      |      | School IT Ticket Category       | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Choice list      |      | School IT Ticket Status         | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Choice list      |      | School IT Ticket Subcategory    | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Table            |      | School IT Ticket                | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Dictionary       |      | School IT Ticket                | admin      | import            | INSERT_OR_UPDATE |
| 2024-09-20 10:17:17 | Dictionary       |      | School IT Ticket Assigned Group | admin      | import            | INSERT_OR_UPDATE |

Figure 3: Completed Project Update Set with captured customer updates and XML export link.

## Deployment & Packaging Architecture

All schema definitions, choice list dependencies, form layouts, Flow Designer actions, and notification triggers created during development were captured inside a dedicated ServiceNow Update Set. Upon verification, the Update Set was marked as Complete and exported as an XML package for version-controlled backup and deployment across environments.